

FACULTY OF COMPUTING**SE2030 - Software Engineering****LAB SHEET 04 - ACTIVITY DIAGRAM****Web-based E-Shopping Store**

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Group Type	Pro-rata / Individual Group
Project Topic	Web-based E-Shopping Store
Main Function	Place an Online Order

This report is prepared for the individual pro-rata project. The single registered student therefore provides the use case scenario and the corresponding activity diagram for the selected main function.

1. Lab Objective

The purpose of this lab is to model the flow of activities within a system by identifying actions, decisions and parallel/alternative paths, and to represent the behavior of a use case using an Activity Diagram.

2. Selected Use Case Scenario

Use Case: Place an Online Order

Primary Actor	Customer
Supporting Components	E-Shopping System, Payment Service, Order/Inventory
Preconditions	Customer can access the store; products are available in the catalogue.
Postconditions	A successful order is stored, stock is reserved/updated, payment status is recorded, and the customer receives confirmation.

3. Main Flow

1. Customer browses or searches for products.
2. Customer selects a product and adds it to the cart.
3. Customer reviews the cart and proceeds to checkout.
4. The system validates the cart and checks stock availability.
5. If stock is available, the system reserves the required stock and calculates the order total.
6. Customer enters delivery details and confirms the order.
7. The system sends the payment request to the payment service.
8. If payment succeeds, the order record is created and the payment status is stored.
9. The system sends the order confirmation to the customer.
10. Customer can view the order and continue to order tracking.

4. Alternative / Exception Flows

- Stock unavailable: the system displays an out-of-stock message and the customer may modify the cart.
- Payment unsuccessful: the system displays a payment failure message and allows the customer to retry or cancel.
- Customer cancels before payment: no confirmed order is created.
- Invalid delivery information: the system requests the customer to correct the required fields before continuing.

5. Activity Diagram with Swimlanes

The following activity diagram separates responsibilities into Customer, E-Shopping System, Payment Service, and Order/Inventory swimlanes. Decision nodes represent stock and payment outcomes, while alternative paths show how the system handles common failures.

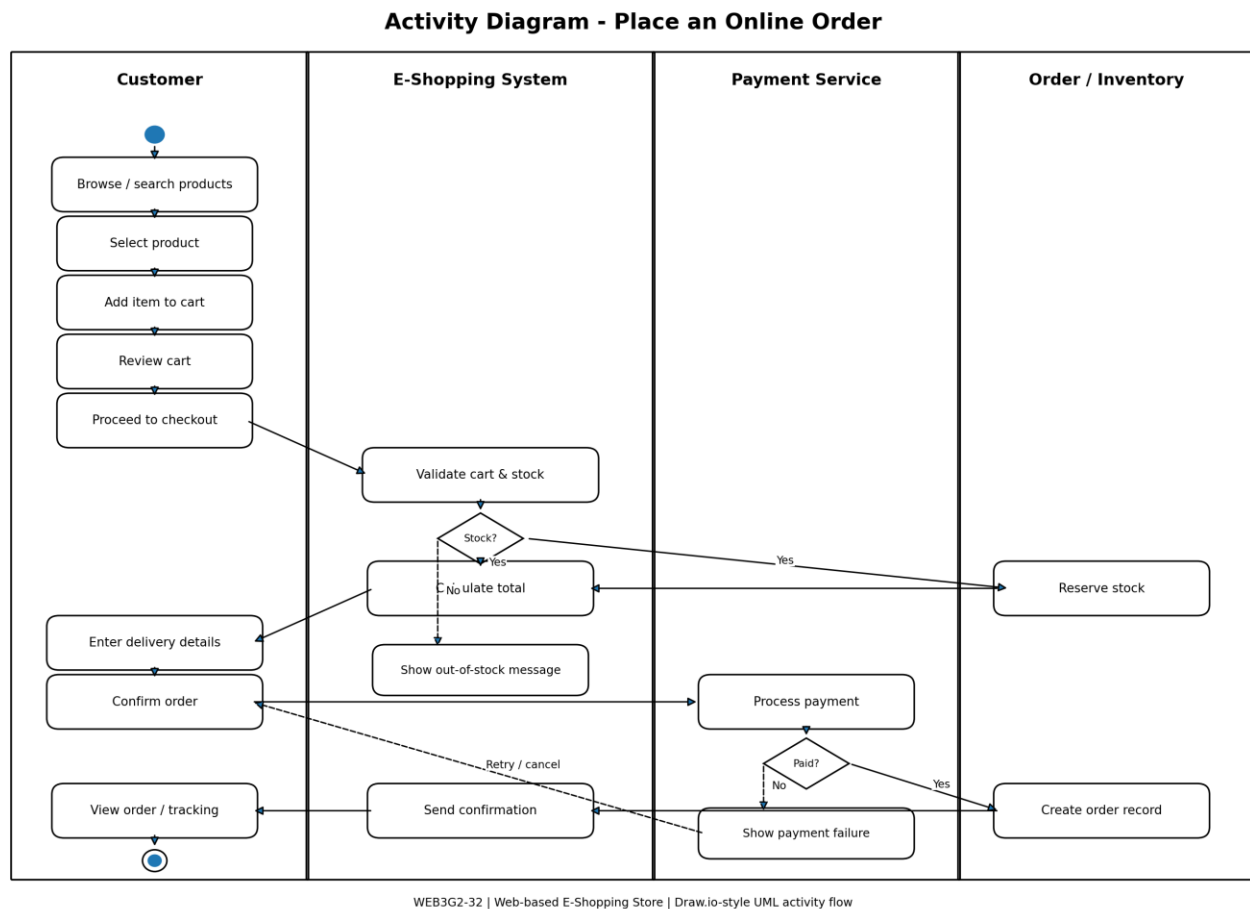


Figure 1: Activity Diagram - Place an Online Order (with swimlanes)

6. Diagram Explanation

The activity begins with the customer selecting products and preparing the cart. Control then moves to the system for stock validation. When stock is available, inventory is reserved and the order total is calculated. After the customer confirms delivery details, payment processing occurs in the payment-service lane. A successful payment leads to order creation and confirmation. Unavailable stock and failed payment are represented as alternative paths, making the diagram consistent with the main and exception flows of the use case.